Micron 126 (2019) 102718

Contents lists available at ScienceDirect

Micron

journal homepage: www.elsevier.com/locate/micron

Automated stitching of microscope images of fluorescence in cells with minimal overlap

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ARTICLE INFO

Keywords:
Image stitching
Brain tumor cell
Tile scan
Fluorescence microscope images
High-content screening

ABSTRACT

The morphology of tumor cells is highly related to their phenotype and activity. To verify the drug response of a brain tumor patient, fluorescence microscope images of drug-treated patient-derived cells in each well are analyzed. Due to the limitation of the field of view (FOV), a large number of small FOVs are acquired to compose one complete microscope well. Here, we propose an automated method for accurately stitching tile-scanned fluorescence microscope images, even with noise and a narrow overlapping region between adjacent fields. The proposed method is based on intensity-based normalized cross-correlation (NCC) and a triangular method-based threshold. The proposed method's quantitative accuracy and the sensitivity of the input was compared to other existing stitching tools, MIST and FijiIS, setting manually stitched images as the ground truth. The test images were 20 samples of 3 × 3 grid images in three versions of the fluorescence channel. The distance between the location of each field and number of cells was determined for different input field overlap ranges (1%, 3%, 5%, and 10%), while the actual value was about 1.15%. The proposed method had a distance error of 1.5 pixels at an input overlap of 1%, showing the lowest minimum error at all channels. Regarding the difference in cell numbers, although the number of overlapping cells was always small because of the narrow overlapping range, the proposed method was able to generate the resultant image with the smallest difference. In addition, to confirm the size limitation of the proposed algorithm, the accuracy of stitching images of grid structures 3 × 3, 5 × 5, 10 × 10–20 × 20 was tested, showing consistent results. In conclusion, quantitative evaluation of the performance of the method proved its improved accuracy compared to other current state-of-art techniques, and it showed robust performance even with noise and a narrow overlapping region between adjacent fields.

1. Introduction

Fluorescence microscope images are used to extract the morphology/geometric features of cancer cells, which are important because they are highly related to the cells' phenotype and activity. For cancer therapeutics, these characteristics have a significant role in prediction of drug response.

Fluorescence microscope images need to be captured at a high resolution to contain a large view while maintaining high spatial resolution. However, due to the limited field of view (FOV), one fluorescence microscope image of cancer cells is composed of a large number of field images with small FOVs (typically nine fields at 10x magnification). To analyze the overall aspect of fluorescence microscope images of cancer cells, tile scanned or panoramic images must be stitched. Image stitching is the process of combining multiple images with overlapping fields of view to produce a panoramic or high-resolution image (Adel et al., 2014).

It is important to stitch multiple field images constituting one well for the following reasons: 1) Drug dosing is administered in the middle of the well. Therefore, no field image could represent the entire well

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https://doi.org/10.1016/j.micron.2019.102718
Received 27 April 2019; Received in revised form 25 July 2019; Accepted 25 July 2019
Available online 16 August 2019
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image pattern. 2) Each field has different boundary conditions, which might affect the pattern/feature of the image. 3) A larger FOV sample is more likely to capture a meaningful pattern than a small FOV sample. 4) Overlap regions contain information about cell existence and cell condition.

Generally, image stitching involves two steps: 1) calculation of the similarity between overlapping regions and 2) merging of field images. The most similar regions between adjacent field images have been found using feature-based and intensity-based methods (Adel et al., 2014; Cho et al., 2015; Park et al., 2014; Rankov et al., 2005; Subramanyam, 2012; Sun et al., 2006; Yang et al., 2013). Feature-based methods find correspondence between image features such as points, lines, and contour lines while intensity-based methods compare the magnitude of changes in the intensity of images to find matching parts.

The feature-based method enables extraction of the feature points of the image to identify similarities. It does not depend on the rotation or size of the image, nor does it rely on a camera's angle of view. Moreover, the method does not require the user to manually define parameters. Sun et al. proposed an algorithm for estimating globally optimal geometric and radiometric transformation parameters for mosaicing microscope images using the Harris corner for detection and matching (Sun et al., 2006). Application of this method to both optical microscopy and fluorescence images can identify corners, assuming that 5% of the minimum image is superimposed. Ma et al. developed AutoStitch, a stitching application that finds matching points on electron microscope images with about 15% overlap using a feature-based registration method called scale-invariant feature transform (SIFT) (Ma et al., 2007). SIFT extracts feature vectors for a local patch centered on each feature point after selecting points that are easy to identify such as corner points. Since local features are sacrificed to some extent for invariant features of scale, rotation, and brightness variations, a number of feature points can be matched to a single feature point.

Both SIFT (Lowe, 1999) and the Harris corner (Harris and Stephens, 1988) are highly accurate. However, the fluorescence microscopy images used in this paper do not require the complex computations associated with these methods because there is no change in size or rotation. Moreover, one cannot expect that enough features will be detected to track the overlapping region when the overlapping range is narrow that the existence of duplicate cells is uncertain. Thus, a brightness-based method with less computation is adopted in this study to increase computational efficiency.

For intensity-based stitching, the mean absolute difference (MAD) or mean square error (MSE) is used as a basic metric for comparing changes in brightness to find similarities between two images (Kang and Ha, 2003). However, since these two metrics are difficult to apply when the overall brightness of the image is not uniform, it is necessary to normalize the brightness value of the comparison area so that the patterns can be compared. In this paper, we adopt normalized cross-correlation (NCC), which is a metric that can simultaneously compare brightness differences between normalized and original images. It measures linear differences and geometric similarities. NCC is one of the typical methods used to find matching points between various medical images such as 3D multiple dental CT, digital X-ray images of the whole body, and optical microscope images of blood vessels.

State-of-the-art automatic stitching algorithms include the Microscopy Image Stitching Tool (MIST) and FijiIS (Chalfoun et al., 2017; Schindelin et al., 2012), both of which were developed for microscopic images. These two algorithms can be applied to fluorescence microscopy images and provide a good image if the user has adequate information about the image and if the image has an adequate overlapping region. Yet, when the image has a very narrow overlapping area and is not in a good condition, e.g., due to noise or blurring, the stitching tools may not be robust and may return incorrectly stitched images. The stitching requires sufficient overlapping area, however, images can be analyzed in global and local due to its goal which means the overall image can be unnecessary. Creating an image with overlapping area takes up unnecessary storage space. If the minimum overlapping area the stitching requires is reduced, the data capacity will be reduced and can also easily approach to one image in global and local view.

Herein, we propose an automated method for seamlessly and accurately stitching tile scanned fluorescence microscope images. A stitched image with minimum error can be obtained by detecting the presence of overlapping cells and finding the most similar regions according to the conditions of the experimental images. We evaluate the performance of our algorithm by comparing it with conventional methods. The experimental results demonstrate the efficacy of the proposed approach. Our algorithm is written in MATLAB (The MathWorks, Inc., MA, USA) and the implemented source code and sample images to be stitched are accessible online (https://github.com/Ewha-Stitching/Stitching.git). The code is freely distributed as an open source tool, and we wrote its pseudo-code to help users understand the proposed method and its code (see Fig. A.1).

2. Material and methods

2.1. Patient recruitments

Fluorescence microscope images contain the drug responses of different types and concentrations of patient-derived cancer cells. The cancer cells were obtained from patients who had undergone brain tumor removal surgery at the Samsung Medical Center (SMC). This study was approved by SMC Institutional Review Board (IRB) (IRB file #201512092).

The cancer cells were obtained from patients who had undergone brain tumor removal surgery at the Samsung Medical Center (SMC), and informed consent was received.

Image data from 42 cases have been acquired and included in this study. The ratio of recruited female patients to male patients is 18 to 24. The mean age of patients was approximately 54 years (± 10.58 years), ranging from 29 to 72. The average age of the female patient group was roughly 56 years (± 9.65 years), and that of the male group was 52 years (± 11.61 years).

All patients have glioblastoma multiforme (GBM) tumors, the most aggressive cancer. GBM cannot be prevented and is generally treated with a combination of chemotherapy, radiation therapy, and surgery. Even after treatment with curative intent, GBM has a high rate of recurrence.

2.2. Tissue preparation and data acquisition protocols

After tissue procurement, mechanical and chemical dissociation as well as purification were performed to separate and cultivate cells from patient-derived tissue. Cultivated cells were treated with seven different concentrations of 60 targeted anticancer drugs on a plate consisting of 384 wells. The 60 drug types were separated into two groups with different drugs listed in each column in two broad rows and different concentrations (μM) listed in each row. After drug treatment, a 2D image of each well was taken at 10x magnification using PerkinElmer's Operetta high-content screening (HCS) system. We acquired images that had a minimal amount of overlapping, as regular analysis in a hospital is performed with minimally overlapping images to exclude duplicated cell feature results in the same well.

To develop an optimal drug response evaluation index by analyzing the responses of patient-derived cells to drug treatment using various biomarkers and readout parameters, the HCS system was introduced for multi-angle analysis based on cell images. HCS can simultaneously evaluate various cell phenotypes and their responses at the cellular level as well as measure the heterogeneity of the cells.

In order to increase the reliability of the experimental results, the experiment was repeated twice for each drug administered to the patient-derived cell. Images were taken in four channels each of which

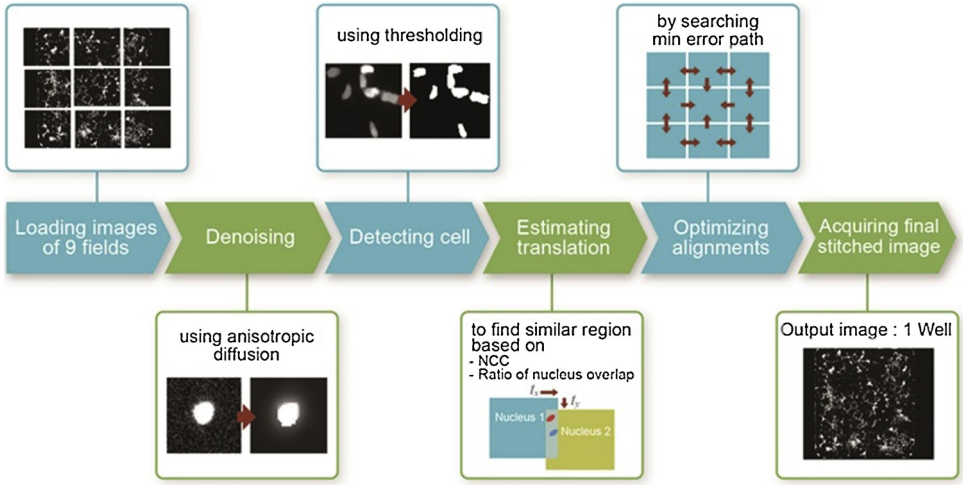

Fig. 1. Pipeline of proposed stitching algorithm for a fluorescence microscope image.

showed apoptosis marker, nuclear fluorescence, proliferation marker, and bright field, respectively. However, in this paper, only the fluorescence images were treated. In the nuclear fluorescence images, the tumor cells were stained with DAPI, a fluorescent blue DNA stain that can pass through an intact cell membrane and stain live and fixed cells. It enables one to count cells and nuclear segments. In the other channel of fluorescence images, the apoptosis marker and proliferation marker each showed apoptosis cells and proliferating cells.

2.3. Proposed stitching algorithm

The proposed algorithm involves four stitching processes, as shown in Fig. 1. De-noising is performed first to enhance the performance of the following detection process which is performed to verify the existence of a cell and the location of the nucleus in the comparison region. Next, an overlap region is identified using NCC and optimal connection to the fixed image is achieved. The matching region will be determined based on intensity, and the comparison region between adjacent fields are gradually enlarged. Lastly, alignments are optimized to find the minimum error path from each field image to one fixed image.

2.3.1. Denoising a fluorescence microscope image

The purpose of the de-noising step is to improve the results of the detection process and find similar regions between neighboring field images. Background noise is often reduced by simply averaging uneven brightness pixels. However, this averaging process mostly focuses on pre-elimination of small amounts of noise with high intensity (e.g., salt-and-pepper noise) to prevent the noise from being recognized as a cell during the detection process and use of incorrect positional information in the matching region process. The brightness of cells at the boundary could be very weak because the overall brightness is weaker farther away from the center of the image. This is likely that the cells with low brightness are considered as noise and thus removed by the averaging process. Therefore, the edges of objects need to be preserved regardless of their intensity while noisy background is suppressed.

Anisotropic diffusion (Perona and Malik, 1990) maintains fine edges better compared with other de-noising methods (Chao et al., 2010). When anisotropic diffusion is applied, the pixel value of an image edge is changed less than other image contents. Therefore, a filter that smooths inside a region but skips the edges of an image is needed. In the current experiment, the denoising process was performed 11 times for each field image. The number of denoising process iterations was determined by adding extra times to the number of the optimum result found in the fluorescent cell image stitching experiment being studied. However, the number of denoising iterations can be unnecessary or more applicable depending on the target image's condition.

2.3.2. Tumor cell detection

The cell detection process determines the presence and position of the nucleus in the overlapping region to estimate the translation between neighboring field images. In this study, a relatively simple and convenient approach of image segmentation for detecting tumor cells using thresholding was applied. Because the brightness of the fluorescence cells in the image becomes weaker near the boundary of the field image and the intensity smoothly transitions from the nucleus to the background, it is difficult to find the optimal threshold and neatly classify cells.

When testing the image with Otsu's method (Otsu, 1979), occasionally some relatively dark nuclei were excluded. Since a lower threshold was required to filter out more structures, the triangle method (Ghosh and Kaabouch, 2016; Zack et al., 1977) was selected. The triangle threshold selects the point at the maximum distance from the diagonal line of the height and range in the histogram, adding fixed offset at the end. By separating the pixels of images into backgrounds and objects, it is also possible to estimate the area of the cell.

2.3.3. Alignment between two field images

To find overlapping regions between two images aligned horizontally or vertically, a region with high similarity to one fixed image is searched while gradually translocating other images in an x, y direction and thus enlarging the overlapping area. The extent of the translation starts from one column to the user-set limit of translation. Fig. 2(a) shows a flowchart of the process of finding the translation in an x, y direction to define the extent of an overlapping region. For each movement, similarity is computed using intensity-based NCC, but only in the region hypothesized to contain a nucleus, which is determined during the previous cell detection step.

The NCC value is calculated on the assumption that the cell exists in the overlapping region of interest, but there is no guarantee of cell existence in that region. If there are no cells in either of the two regions being compared, translation is done over all overlapping regions of the background with the aim of finding the most similar region. In such a case, the correlation value is manipulated by adding a negative value (-2) to maintain a lower level of similarity than that of a region containing nuclei.

Given that t_x and t_y are the translation in the x- and y- directions in Fig. 2(b), respectively, t_{x_0} is the first column-wise translation where the nucleus first appears in the overlap region. In addition, $t_{x_{end}}$ is the user-set limit of translation in the x-direction. In this paper, ± 2% of the

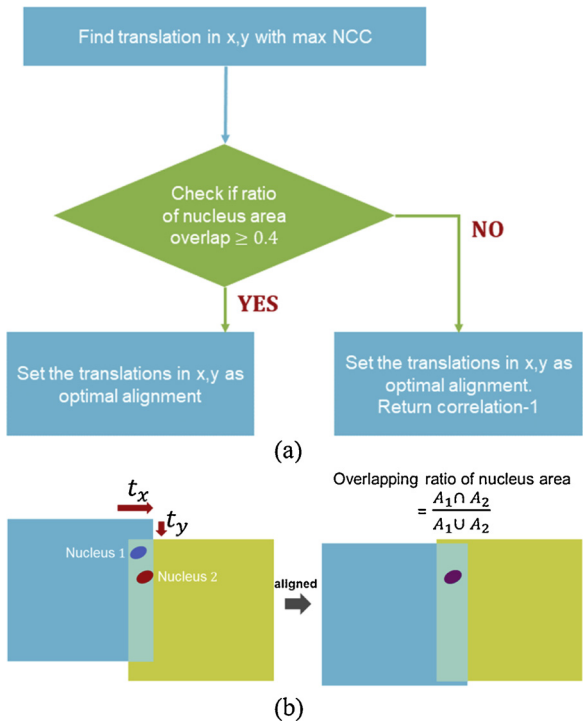

Fig. 2. The estimation procedure of field image translation. (a) the flowchart of the process of finding translation in an x, y direction. (b) the method of checking the ratio of the nucleus areas, A1 and A2, in the overlap between images 1 and 2, respectively.

image width and height is used as the limit of translation in the x- and y-directions. A detailed pseudo-code of the translation procedure is given in S1 Fig (see ***function*** TRANSLATION).

The NCC-based similarity between image i (imi) and image j (imj) is computed as follows:

$$\mathrm{NCC}_{i,j} = \frac{\sum_{x=0}^{M}\sum_{y=0}^{N}(im_i(x,y)-\bar{im_i})(im_j(x,y)-\bar{im_j})}{\sqrt{\sum_{x=0}^{M}\sum_{y=0}^{N}(im_i(x,y)-\bar{im_i})^2}\sqrt{\sum_{x=0}^{M}\sum_{y=0}^{N}(im_j(x,y)-\bar{im_j})^2}} \quad (1)$$

where $im_i(x, y)$ represents pixel intensity at position (x, y) in im_i and $\bar{im_i}$ is used for the mean of pixel intensity, and t_x and t_y are given as a constant M, N (Kang and Ha, 2003).

Since normalization is performed in Eq. (1), it is possible to compare the intensity patterns of two images regardless of differences in their background brightness. The value of correlation ranges from 0 to 1 representing the highest similarity. By gradually changing the overlapping region, the region with the highest similarity are identified and used for inter-field alignment. However, regardless of the correlation, one should consider that a region with high correlation could contain different objects. The region is considered to be matched when the overlapping ratio of the nucleus area is 0.4 or more. If the ratio is less than 0.4, irrespective of the correlation value, the contents are considered to be different. In this case, the field image is translated to the middle of a background that can be overlapped, and a negative value (-1) is added to the correlation, similar to what is done when an image pair has no objects in the region of interest.

2.3.4. Alignment optimization across entire field images

The translational alignment between two field images is obtained in the previous step, and the correlation value between each neighboring

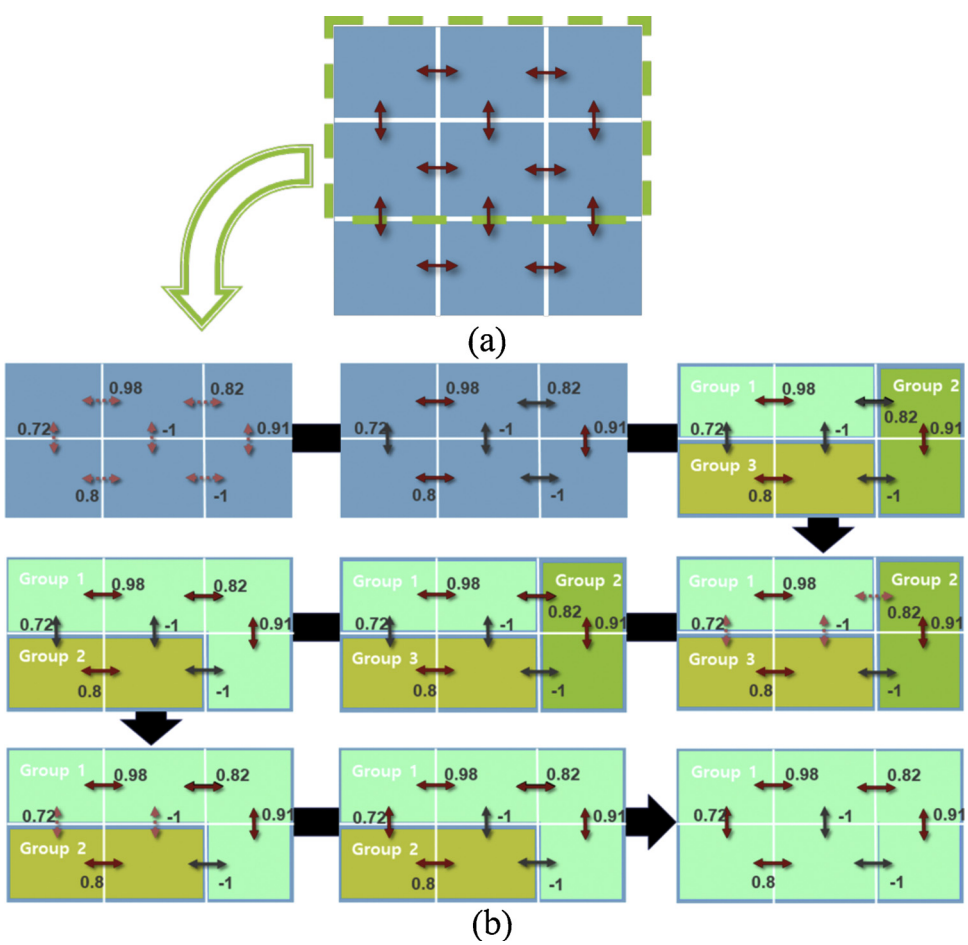

Fig. 3. Arrangement of field images. (a) Possible connections between field images. (b) An example of optimal alignments of six fields, with one fixed field. A dashed arrow indicates a possible path in the field of interest to be aligned (highlighted), and the colored arrow indicates the path selected based on the NCC value. The number on the bi-directional arrow is the NCC value between adjacent field images.

image represents the trustworthiness of the estimated translation. In this step, we optimize alignment across entire fields using the correlation information between two fields.

Fig. 3 presents the representative procedure for optimal alignments of upper rows in the sample well. All field images start with finding the highest correlation to the neighboring field in the link and then creating a group. If the number of groups is more than two, it means that all the fields are not connected in one. Searching for the highest correlation link between groups and then merging the groups is done repeatedly until only one group is left.

Each field is associated with at least two other fields. The fields are structured in a 3 × 3 grid, as shown in Fig. 3(a). This means that one misconnection may cause dislocation error which is accumulated as alignment between images is performed along the path connecting the field to be aligned to the fixed center field. Therefore, it is important to choose the path with the lowest error. To find the optimal path, we selected the path between possible routes based on the value of the correlation with maximum value. A detailed pseudo-code of the procedure of computing the minimum correlation is given in S1 Fig (see function OPTIMAL_PATH).

In the end, the global position of each field image is obtained by integrating the relative position between the center and the field images. According to the global position of each field, images are merged into one extracting maximum intensity in overlapping areas.

We mainly used a well consisting of nine fields in a 3 × 3 grid to evaluate our method. However, as long as the field images are in a grid pattern, our method can be easily applied to images that are positioned in other grid patterns (e.g., 2 × 2, 3 × 2, 10 × 10, etc.). To demonstrate that our algorithm can be applied to all size of grid, we also tested the performance on maximum 20 × 20 grid pattern images.

2.4. Quantitative accuracy measurement

To determine the accuracy of the proposed method, its performance is compared to that of other stitching support tools: MIST (Chalfoun et al., 2017), FijiIS (Schindelin et al., 2012). As shown in Fig. 4, 20 wells of various types were selected based on the amount and distribution of cells. Some have uniformly distributed cells throughout the wells, while others are concentrated to a specific side. Images may include cells that range from crumpled to scattered and from massive to little. Among the twenty wells, some do not have redundant cells in the overlap region. The images are of three versions of the fluorescence channel: 1) the apoptosis cell, 2) the nucleus, and 3) the proliferating cell. The shape and brightness of fluorescence take on different aspects in each channel, as shown in Fig. 5.

Considering manually stitched images to be the ground truth, two metrics were used to determine quantitative accuracy: 1) 2D pixel-wise distance between the manually stitched image and stitched image using each method, including the proposed method, and 2) the difference in the number of cells between the reference and image obtained after employing each method.

It is very important to accurately count the number of cells within a given area of a sample (i.e., the concentration of cells within a sample) in medical diagnosis and treatment. Moreover, the number of cells is a useful indicator in measuring cell viability. Counting the number of both live and dead cells in a cell culture is important for determining the life expectancy of a cancerous cell, the effectiveness of anticancer drugs in the treatment of malignant, or cancerous disease. Using the second metric related to the number of cells, we analyzed a cell count error caused by misalignment of the field images in a well. The number of cells were counted in the resulting stitched images using the open source CellProfiler software (Lamprecht et al., 2007).

Information about the sequence of the field image acquisition and approximate overlap range (Fig. 6) are required in advance before implementing the stitching method. Additionally, FijiIS and MIST ask a user to enter more specific inputs including threshold level and overlap uncertainty. Unlike the proposed method, the performance of the tools was sensitive to changes in the input values, and their performance was poor without reasonable estimates of the inputs. For a fair comparison between the proposed method and the current methods, the input values which derived the optimal performance of the current methods were manually searched and then the image stitching was performed.

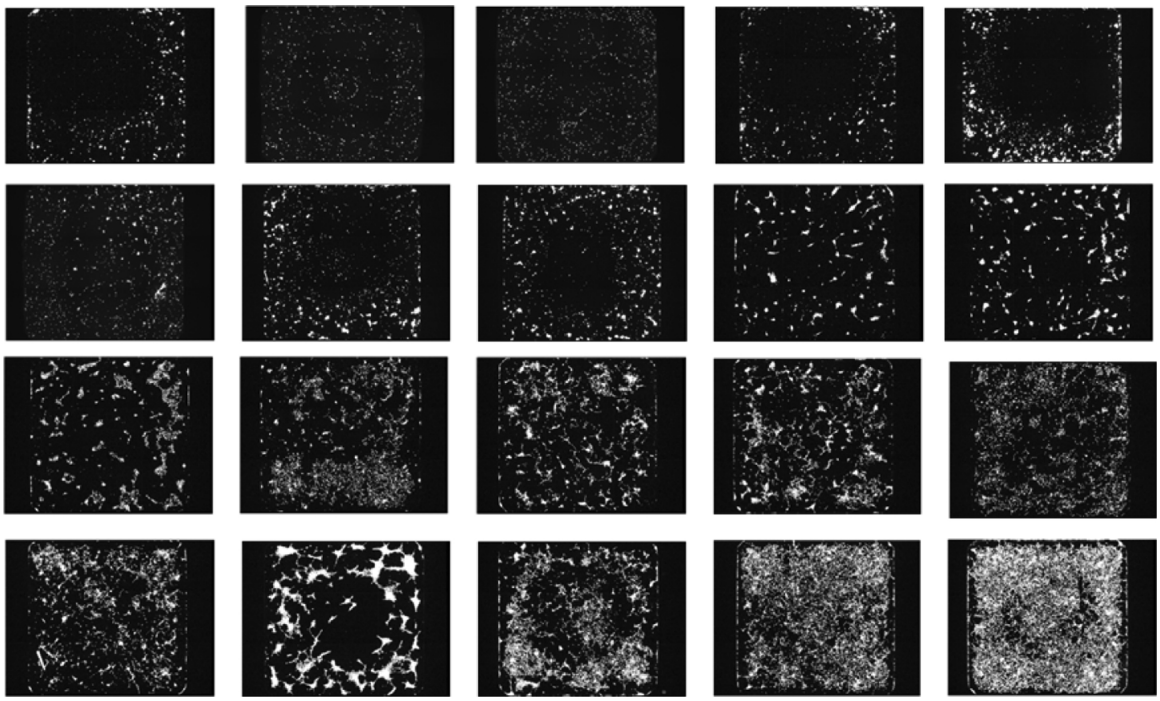

Fig. 4. Twenty representative wells consisting of nine field images were manually stitched to generate reference data for evaluation. The resolution of each field image is 1360 × 1024 pixels.

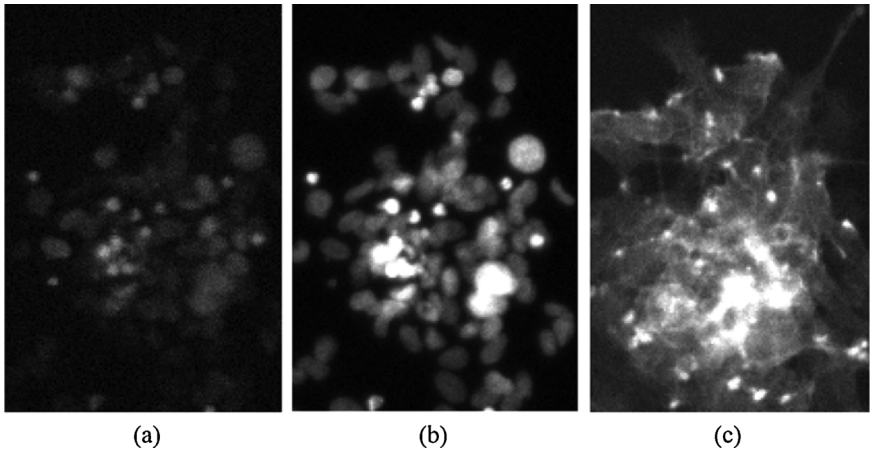

Fig. 5. Glioblastoma cells in three channels: (a) apoptosis, (b) nucleus, and (c) proliferating.

3. Results

3.1. Stitching accuracy

Table 1 and Fig. 7 show the average distance of the field images for each input overlap based on 20 sample wells (1%, 3%, 5%, and 10%) consisting of 3 × 3 field images. The uncertainty of the overlap is set equal to the input overlap, so the search range always contains the actual overlapping area. When the overlap range is close to the actual range of the image, MIST measures a considerably small error distance. However, the result varies greatly depending on the input value. The proposed method has consistent outcomes unless the entered overlapping range is smaller than the actual range by considering whole possibility of overlap. Therefore, unless there is no overlapping region contained in the input overlap, it identifies the closest position to the manually stitched image. Moreover, the performance/accuracy patterns over the different sizes of grids is described in Fig. 8 using three cases of stem cell colonies' images. The results show strong similarities between small 3 × 3 images and large 20 × 20 images, with the remaining fluctuation of distance error under 1.06 pixels

Fig. 9 shows the average difference between the number of cells in the manually stitched images (i.e., reference) and that in the stitched images by different stitching algorithm including the proposed method.

Table 1
Average distance error (pixel) for each input overlap.

Overlap (%)	Tool (pixels)		
	MIST	Fiji	Proposed
1%	1.65	33.61	0.88
3%	8.27	18.11	0.88
5%	32.71	70.55	0.88
10%	81.94	167.57	0.88

MIST, FijiIS, the proposed method, and the stitching interface without an overlapping area were used to compare manually stitched images. MIST and the proposed method showed a relatively small difference in cell number compared to the other methods. At an input overlap of 1%, the proposed method achieved the minimum difference in cell number, meaning that it successfully merged the most duplicated cells which would be displayed separately in mis-stitched images although they are the same cell.

3.2. Field-wise comparison of cell features

The importance of stitching is determined by whether a feature

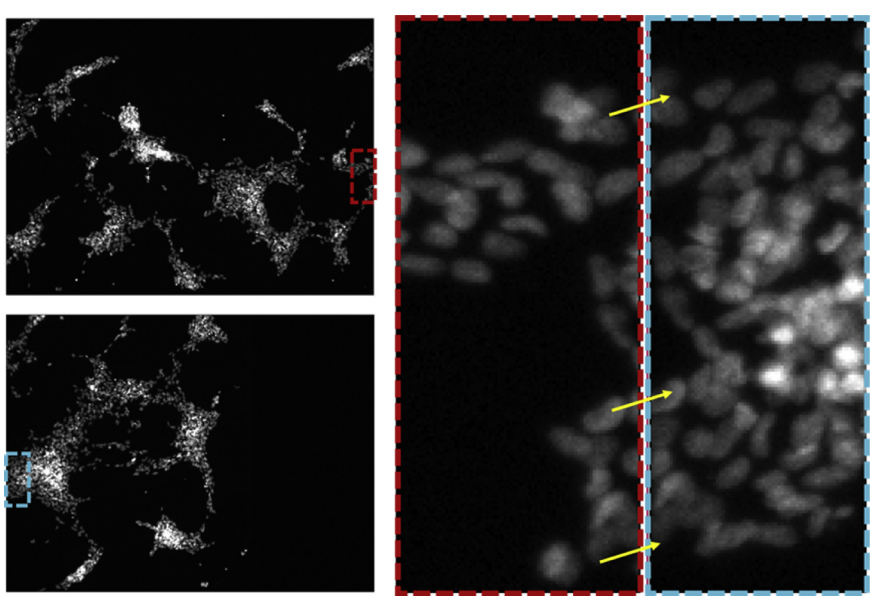

Fig. 6. Overlapping region between two neighboring fields in a horizontal direction at the channel 2.

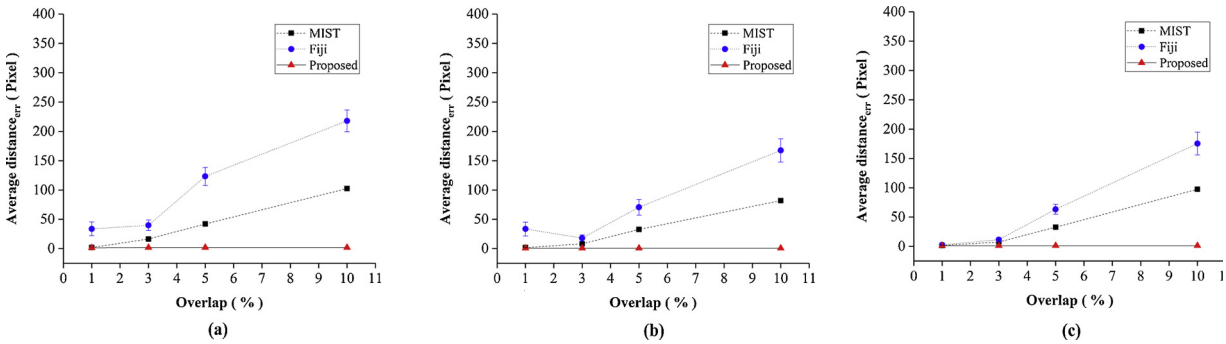

Fig. 7. Average distance error for each input overlap of stitched well images at (a) Channel 1 (b) Channel 2, and (c) Channel 3. Each of the representative twenty wells has nine field images in a 3 × 3 grid.

extracted from a random field can represent the state of one well. Various features, such as area, intensity, roundness, and the distance between cells, are extracted to identify the patient's drug response. Some features may be meaningful, and the applicability of some are not yet determined. It does not matter if a meaningless feature differs between an unstitched image and a stitched image. However, to identify meaningful features, precise analysis must be performed, which means that every feature must present values in a well unit. This again leads to the previous question: can a feature extracted from a random field represent the state of one well? S2 Fig presents the cell-present pixel area of each field image with five representative wells. There is a relatively significant difference between the fields, revealing that a single field cannot represent a well.

4. Discussion

This study proposed a novel stitching method capable of automatically stitching field images, even with a narrow overlapping region (down to 1%) and noise. Regardless of the presence of cells in the overlapping region, the method performed well compared to the other methods (MIST, FijiIS, and AutoStitch). Moreover, the results of this study are a first attempt at morphologically analyzing brain tumor cells in stitched microscopy images taken from a live human brain.

The proposed method showed consistently good accuracy under various conditions, as shown in Table 1 and Fig. 7. Reliably good performance can be achieved by employing the following pre-processing procedures: 1) anisotropic diffusion, which prevents noise from being recognized as a nucleus in the detection process, 2) exclusion of blank backgrounds and determining a threshold based on the triangle algorithm, which allows all cells, including very dark cells, to be used to find the optimal alignment between field images, and 3) determination of optimal alignment based on the NCC-based similarity computation and the ratio of the nucleus as well as overlap between field images.

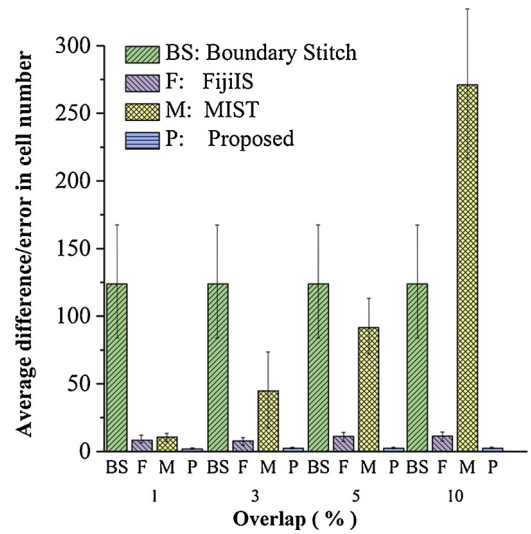

Fig. 9. Average difference/error in cell number between stitched wells using the proposed algorithm and other comparison tools and manually stitched wells (i.e., reference) at various input overlaps.

Here, other automatic stitching algorithms, the performance of which was compared with ours, generally expected the clear existence of overlapping objects in the overlap area. Such algorithms might be unsuitable for fluorescence imaging systems with narrow overlapping areas. For instance, our system has a maximum overlap range of approximately 2.6%. Considering mechanical stage repositioning error (i.e., the overlap error) due to a pre-defined actuator backlash, the overlap range could be even less than 1%. Given the results shown in

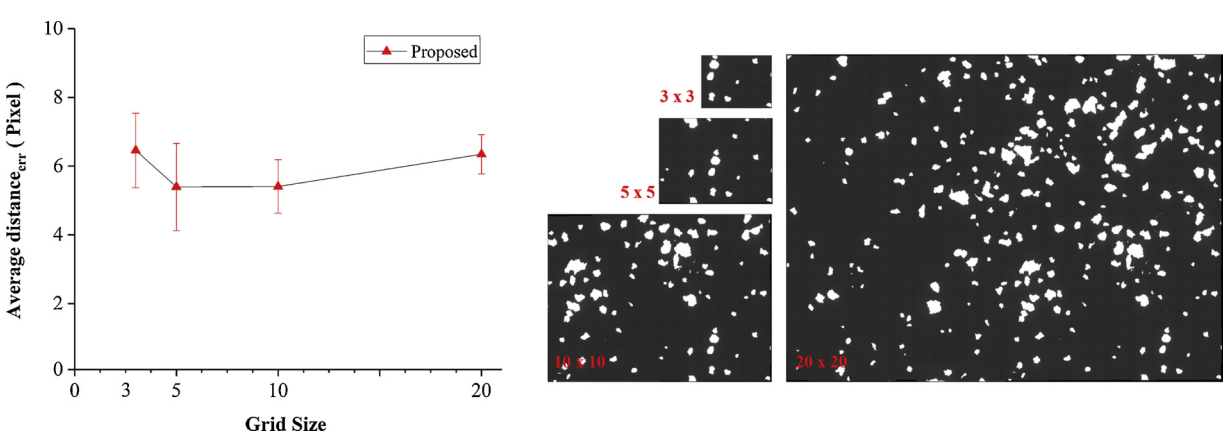

Fig. 8. Average distance error of proposed algorithm for various grid sizes and a sample stitched image. One pixel occupies $0.70*10^{-6}$ % of the image field.

Fig. 7, the proposed method can be considered more reliable than other methods for stitching images with insufficient overlap area and information. Moreover, it has been demonstrated that our algorithm can be applied to various grid size. The open source codes of the proposed methods are well compatible with all common grid patterns of the field images.

This study proved the need for stitching based on quantitative analysis. The manual stitching process is time-consuming and thus is not pragmatic. For that reason, physicians often rely on the results extracted from images simply stitched along the boundary of the field images (i.e., boundary stitch in Fig. 1) or from individual field images that did not undergo stitching. However, as boundary stitched images shows considerable differences in cell number compared to those that were accurately stitched using the proposed method (see Fig. 9), key features extracted from boundary stitched images could be inaccurate. Moreover, field image-based analysis of key feature values obtained without stitching could be inaccurate. S2 Fig shows that each field has different values of the cell-present area feature depending on its position in the grid of a well, and thus a single field image cannot properly represent the entire well. This field-wise difference might occur because drug dosing is performed at the center of the well and each field image has a different boundary condition depending on its position in the grid pattern of the well. The cell-present area is a representative feature of the various features often extracted from fluorescence microscope images. Other features may exhibit a more extreme difference, such as features related to neighboring fields.

The limitations of the proposed method need to be discussed. First, it was assumed that the mechanical stage of a microscope has only translational movement while it positions the sample in a repeatable grid pattern in the field of view. Image rotation was not considered as its change is negligible relative to that of image translation. We could improve the accuracy of our method by taking into account small rotational movements caused by vibration of the mechanical stage. Second, the proposed, NCC-based method can handle image stitching with superior accuracy with a narrow overlap, compared to other methods in comparison, at the cost of longer runtimes. The proposed algorithm experienced, that is, the longer the search time became linearly, the larger the search range. However, similarity checking for the overlapping regions can be processed in parallel, thus if GPU implemented as in MIST, the computation time could be dramatically reduced.

Future work should investigate how stitching accuracy affects the values of the extracted key features and drug responses in cancer cells.

5. Conclusion

In this paper, we propose an automated method for accurate stitching of tile scanned fluorescence microscope images. The method uses intensity-based NCC to find matching regions between neighboring field images and optimally align the field images in a grid pattern in a well. Quantitative evaluation of the performance of the method shows that it is more accurate than other methods, even with narrow overlapping regions between the field images and regardless of the presence of cells in the overlapping region.

Funding

This research was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government, MSIP (grant no: NRF-2015M3A9A7029725 and NRF-2017M2A2A6A02070522, URL: http://nrf.re.kr) and the Ewha Womans University Research Grant of 2017. The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Declaration of Competing Interest

The authors have declared that no competing interests exist.

Acknowledgements

We thank Institute for Refractory Cancer Research, Samsung Medical Center for providing microscopic images of glioblastoma cells. We also appreciate the National Institute of Standards and Technology for providing us with the stem cell colonies' images.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.micron.2019.102718.

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